

digit SOC fixed effects, a one-standard-deviation increase in seniority is associated with a 3.68 percentage point decline in displacement intensity ($p = 0.002$). These patterns are consistent with the framework’s assumption that junior-type tasks are disproportionately the ones GenAI can automate. Supplemental Appendix A.7 provides full details on the data, methodology, and the complete set of results.

5.2 Employment Dynamics by Adoption and Seniority

5.2.1 Descriptive Employment Trends

We begin by comparing the evolution of junior and senior employment over time in GenAI-adopting versus non-adopting firms. Figure 4a plots average junior employment across the two groups. From 2018 through the end of 2022, junior employment in adopting and non-adopting firms followed closely parallel trajectories. Starting in late 2022, however, the trends diverged sharply: junior employment in adopting firms began to decline markedly, while employment in non-adopting firms remained relatively stable. By contrast, Figure 4b shows that since 2018, senior employment in adopting and non-adopting firms grew at a similar pace—aside from a small and brief acceleration in adopting firms in 2021—with no apparent break in trend around 2023. These patterns are consistent with Brynjolfsson et al. (2025a), who document a similar divergence by occupational exposure among younger workers in U.S. payroll data, with no corresponding shift for older workers.

Appendix A.17 formalizes these descriptive patterns with a difference-in-differences specification that compares employment in adopting and non-adopting firms, estimated separately for juniors and seniors. The estimates confirm the visual evidence: junior employment at adopters is flat relative to non-adopters through 2022Q4 and then turns sharply negative, falling by about 9 percent six quarters after the diffusion of GenAI.

5.2.2 Triple-Difference by Seniority

To directly assess the “seniority-biased” effects associated with GenAI adoption, we estimate a triple-difference specification that compares changes in junior versus senior employment within adopting firms relative to non-adopters:

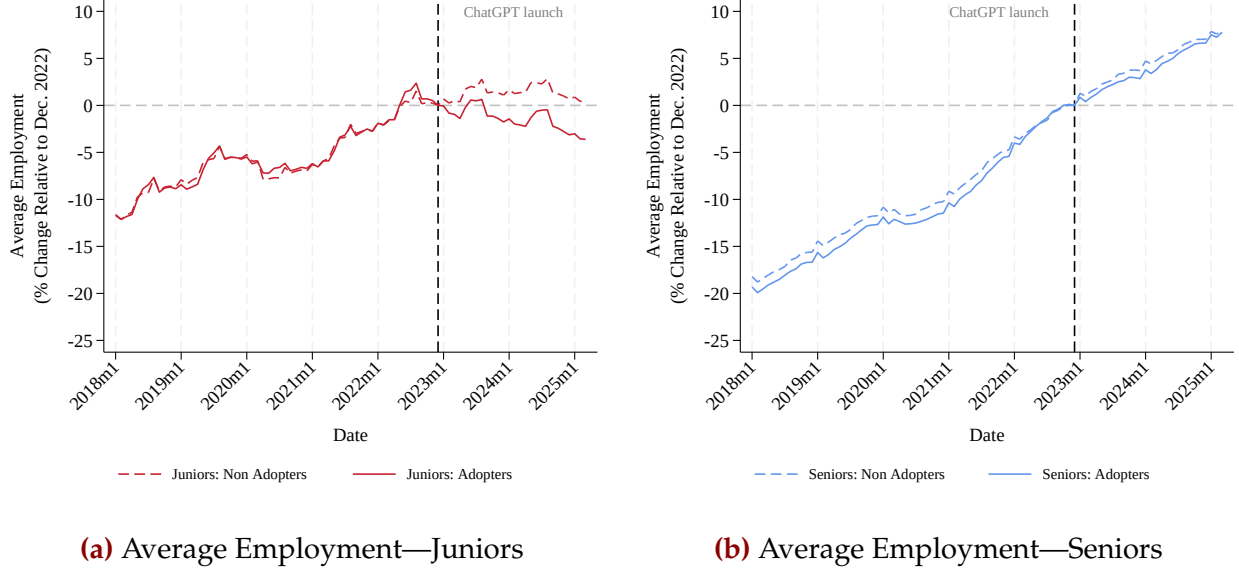


Figure 4: Average Junior and Senior Employment by Adoption Status

Notes: Panels (a) and (b) show the average number of junior and senior workers, respectively, in adopting versus non-adopting firms (as percentage change relative to December 2022, immediately following the launch of ChatGPT). The adoption measure used in this analysis is time-invariant; hence, the figures simply compare adopters and non-adopters. For results using a time-varying adoption measure, see Section 5.3.

$$\begin{aligned}
\log(\text{Employment}_{ist}) = & \alpha + \sum_{j=2018Q1}^{2025Q1} \beta_j \mathbf{1}\{t = j\} \times \text{Adopt}_i \times \text{Junior}_s \\
& + \sum_{j=2018Q1}^{2025Q1} \pi_j \mathbf{1}\{t = j\} \times \text{Adopt}_i + \sum_{j=2018Q1}^{2025Q1} \rho_j \mathbf{1}\{t = j\} \times \text{Junior}_s \\
& + \kappa (\text{Adopt}_i \times \text{Junior}_s) + \gamma_{it} + \xi_{p(i)st} + \varepsilon_{ist}.
\end{aligned} \tag{11}$$

Here, $\log(\text{Employment}_{ist})$ denotes the log employment of workers in seniority group $s \in \{\text{junior}, \text{senior}\}$ at firm i in period t . The indicator $\mathbf{1}\{t = j\}$ equals one in period j and zero otherwise. Adopt_i is a firm-level dummy equal to one for firms that adopt GenAI, and Junior_s is an indicator equal to one for juniors and zero for seniors. $p(i)$ denotes the sector (NAICS 2 digit) of the firm i .

The coefficients β_j trace a triple-differences event-time profile: they capture how junior employment evolves relative to senior employment *within the same firm and period*,

comparing adopters to non-adopters. Firm-by-time fixed effects γ_{it} absorb any shocks or trajectories specific to a given firm in a given period. The industry-by-seniority-by-time fixed effects $\xi_{p(i)st}$ control for broad sectoral trends that differentially affect juniors and seniors over time, ensuring that our results are not driven by sector-wide shifts in junior versus senior employment.

Figure 5 presents the estimated coefficients β_j from Equation 11. Estimation begins in 2018Q1 due to computational constraints. The coefficients remain essentially flat through 2022Q4, aside from a brief dip in early 2021.²⁶ Beginning in 2023Q1, the coefficients drop sharply, reaching roughly a 9 percent decrease after six quarters. This break—coinciding with the rapid diffusion of GenAI—provides suggestive evidence that adoption is associated with increasingly seniority-biased patterns, reducing junior employment relative to senior employment within firms.²⁷

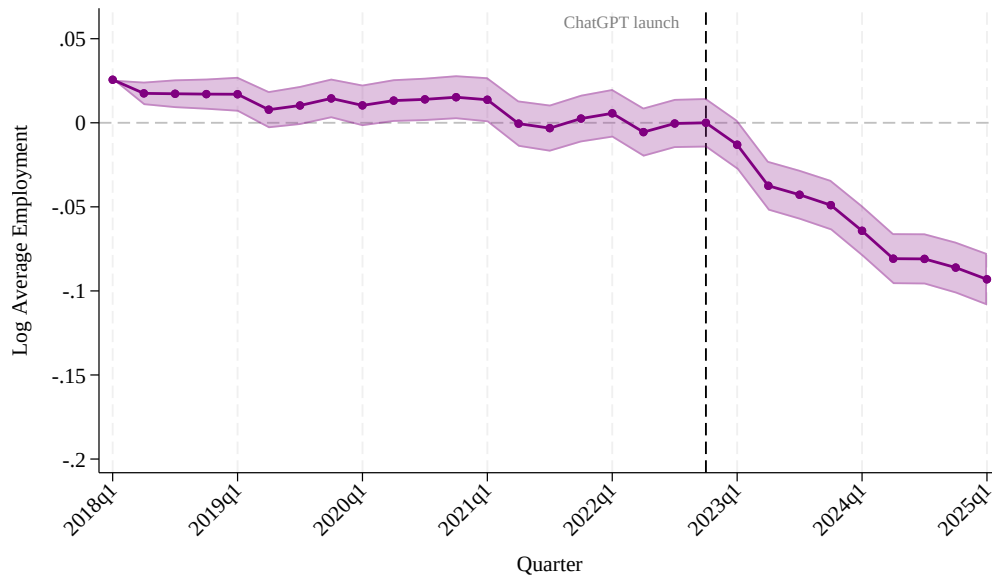


Figure 5: Triple-Difference Estimates: Juniors vs. Seniors

Notes: The figure presents the estimated triple-difference coefficients β_j from Equation 11, capturing how junior employment evolves relative to senior employment within the same firm and period, comparing adopters to non-adopters. Standard errors are clustered at the firm level.

The β_j coefficients can be interpreted as reflecting the differential impact of GenAI

²⁶The patterns shown in Panels 4a and 4b indicate that this temporary decline likely reflects a brief acceleration in senior employment among adopting firms in 2021, rather than a decrease in junior employment.

²⁷In Supplemental Appendix A.18, we present triple-difference estimates excluding the industry-by-seniority-by-time fixed effects. The results are very similar to those in Figure 5.

adoption on junior relative to senior employment, provided that—after conditioning on firm-by-time and industry-by-seniority-by-time effects—no other factors since 2023 systematically affected juniors and seniors differently across adopting and non-adopting firms. This might be a concern due to the potential endogeneity of GenAI adoption. The close similarity in relative junior employment dynamics between adopters and non-adopters from 2018 through 2022 helps alleviate this concern by supporting the plausibility of parallel pre-trends. We provide further support for our interpretation through analyses by occupational exposure (Section 5.2.3) and a staggered event-study design (Section 5.3). Section 5.4 further discusses possible forces that could challenge this interpretation and how we address them.

5.2.3 Triple-Difference by Occupational Exposure

The previous section showed that GenAI adoption is associated with a decline in junior employment relative to senior employment. We now examine whether this seniority-biased effect is concentrated in occupations most exposed to GenAI. To do so, we extend the triple-difference specification of Equation 11 by replacing the Junior_s dimension with an indicator for *high-exposure* occupations, and by estimating the specification separately for juniors and seniors. This yields, for each seniority group, a comparison of how employment in high-exposure occupations evolves relative to low-exposure occupations, between adopters and non-adopters.²⁸ Specifically, for each seniority group $s \in \{\text{junior}, \text{senior}\}$ we estimate:

$$\begin{aligned} \log(\text{Employment}_{ict}^s) = & \alpha + \sum_{j=2018Q1}^{2025Q1} \beta_j^s \mathbf{1}\{t = j\} \times \text{Adopt}_i \times \text{HighExp}_c \\ & + \sum_{j=2018Q1}^{2025Q1} \pi_j \mathbf{1}\{t = j\} \times \text{Adopt}_i + \sum_{j=2018Q1}^{2025Q1} \rho_j \mathbf{1}\{t = j\} \times \text{HighExp}_c \\ & + \kappa (\text{Adopt}_i \times \text{HighExp}_c) + \gamma_{it} + \xi_{p(i)ct} + \varepsilon_{ict}, \end{aligned} \quad (12)$$

where $\log(\text{Employment}_{ict}^s)$ denotes log employment of seniority group s in exposure cell $c \in \{\text{high}, \text{low}\}$ at firm i in quarter t ; HighExp_c equals one for high-exposure occupa-

²⁸Supplemental Appendix A.22 shows the most common high- and low-exposure occupations by industry, ranked by number of positions.